00-M-1688
November 3, 1972
(TO SUPERSEDE IN PART)
Int. Fed. Spec. 00-M-00681c
April 30, 1965, and
Fed. Spec. 00-M-681a
August 30, 1955

FEDERAL SPECIFICATION

MOWERS, LAWN, POWER (ROTATING REEL, GASOLINE ENGINE DRIVEN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heavy duty, industrial, selfpropelled, walk-behind, rotating reel lawn mowers with gasoline-engine driven cutting blades.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Sizes. The power mowers covered by the specification shall be of the rotating reel, selfpropelled variety and furnished in the following sizes as specified herein and in the invitation for bids, contract, or order (see 6.2):

Size A - 21 inch nominal.
Size B - 30 inch nominal.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, for Packing.
UU-T-81 - Tags, Shipping and Stock.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose.
MIL-E-10062 - Engines, Spare or Installed, Other than Aircraft, Preparation for Shipment and Storage of.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine Preservative.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications; Electronic, and Associated Electrical Equipment).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI), Inc., Standard:

B71.1 - Safety Specification for Power Lawn Mowers.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106. 222 South Riverside Plaza, Chicago, IL 60606.)

Society of Automotive Engineers (SAE) Standards:

J551 - Measurement of Electromagnetic Radiation from Motor Vehicles (20 - 1,000 MHz).
J607 - Small Stock Two Stroke and Four Stroke Cycle Gasoline Engine Test Code.

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Condition. Lawn mowers shall be complete and new. Handles need not be affixed to the machine. There shall be no defects or noticeable wear in any of the component parts of mechanisms upon the conclusion of any tests conducted to determine compliance with the service, cutting, or other performance requirements of this specification. Mowers shall be free from rust and imperfections which may affect appearance or serviceability of the mower.

3.1.1 Standard product. The lawn mowers furnished under this specification shall be the manufacturer's current standard, industrial product except for those modifications or changes required in the standard product in order to comply with the requirements of this specification. All accessories and components shall be furnished with each lawn mower that are normally furnished with the standard product offered commercially.

3.2 Materials. Materials used in the construction of lawn mowers shall be of the highest grade and suitable for the purpose intended.

3.3 Fasteners. Bolt and nut threads shall conform to H28. Swaging, peening, or staking of threads shall not be permitted. Sheet metal screws and sheet metal nuts shall not be used, unless approved by the procuring activity. Where aluminum-magnesium or zinc alloys are involved, assembly of parts shall be by the following: through bolts, bushings with tapped holes, threaded studs inserted to a depth of two stud diameters minimum, coil type inserts, machine screws in holes tapped directly in the metal where frequent disassembly is not required.

3.4 Bid sample. When specified (see 6.2), and prior to the award of the contract, the supplier shall furnish the mower(s) to prove that his production methods and choice of design detail will produce lawn mowers that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the bid sample during production shall be subject to the approval of the contracting officer. Approval of the bid sample by the contracting agency shall not relieve the supplier of his obligation to furnish lawn mowers conforming to this specification.

3.5 Design and construction. The mowers shall consist essentially of a gasoline engine, blades, guards, controls, handle and propulsion unit. Unless otherwise specified (see 6.2), sizes within a range of plus 2 inches to minus 1 inch of the nominal size specified will be accepted. Mowers shall have a cylindrical rotating reel and a fixed bed knife. The ends of the cutting reel shaft shall be supported at the side frames. The side frames shall, either directly or indirectly, form the main structural supports for the mounting members of the engine, power drive, and traction drive. Mowers shall conform to the safety requirements and shall be capable of passing the tests as specified in the ANSI B71.1 without sustaining damage that could be hazardous to the operator. Mowers shall be capable of operating continuously eight hours daily for sixty days under the cutting conditions and at the operating speed herein specified

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without the necessity to replace or recondition the piston, piston rings, valves, or cylinder bore or involving repair to the differential or clutch. Frames and structural supporting members shall be either heavy or reinforced construction, and shall be capable of withstanding loads arising from operation on rough grounds without breakage or visible deformation. Components and related parts shall be accurately and permanently aligned. Assembled machines shall have adequate rigidity and shall not vibrate objectionably when in operation. All bearing housings shall adequately protect the bearings against the entrance of grass clippings, dirt, and other foreign material. All mowers shall have at least one forward speed and size B mowers shall have one reverse speed.

3.5.1 Guards. Effective guards shall be provided in accordance with ANSI B71.1 to overcome hazards created by moving parts or exposed parts heated during operation of the mower which could cause injury to the operator or others.

3.5.2 Cutting reel. When an object becomes wedged between a reel blade and the bed knife, no material damage to the mower shall occur and power to the reel drive shall be discontinued either through belt slippage, if belt drive, or engine stall. The cutting reel shall be of rugged and durable construction and shall be accurately ground cylindrical with respect to the shaft. Not less than two reel blades shall be in contact with the bed knife at all times. Mowers shall have a frequency of cut such that not less than 9 cuts are made by the reel in a one-foot linear travel distance of the mower (see 4.7.2). The spiders shall be either malleable iron or steel. Each reel blade shall be attached securely to each spider by either riveting or welding. The cutting reel blades shall be either alloy or high carbon tool steel through hardened and shall be Rockwell hardness C 37 to 47. The cutting reel shaft shall be supported on roller or ball bearings.

3.5.3 Bed knife. The bed knife shall be either alloy or high carbon tool steel through hardened with a minimum hardness of Rockwell C 47. In all cases, the bed knife shall be harder than the reel blade when measured within 3/16 inch of the cutting edge after the knife is finish-ground. The portion of the surface of the backing bar making contact with the bed knife shall be finished (rough ground) to a plane surface. Either the backing bar or the cutting reel shall have a screw mechanism for setting proper clearance between the reel and bed knife and shall remain securely locked and shall not creep or loosen under any operation conditions of the lawn mower. The screw mechanism shall be easily accessible so that adjusting tools will not slip out of screw slots or wrenches off of screw heads.

3.5.4 Engine. The engine shall be single cylinder and shall operate in four cycles using gasoline as fuel. The engines shall be equipped with a fuel strainer, a dry element or polyurethane air filter, an efficient exhaust muffler or silencer that directs the outflow to the right or left, forward and down, or to the rear away from the operator, and a starter of the recoil type that automatically releases when the engine starts. Recoil starters shall be mounted with removable threaded fasteners, rather than with rivets to facilitate disassembly. Magnetos shall be the high-tension type and moisture resistant. Magneto points shall be protected against interference by, or accumulation of, grass clippings, dirt particles, and oil. Exhaust valves for engines shall be hard faced heat resistant alloy. Cast iron cylinder block valve seats shall be replaceable. The engine crankshaft shall be not less than 25 millimeters (0.9843 inch) in diameter throughout its length and shall be enclosed in an oil tight housing supported (top and bottom or both ends) on ball or roller bearings or metal alloy bushings. The rated brake horsepower shall be 2.5 horsepower minimum for size A mowers and 4.5 horsepower minimum for size B mowers when tested in accordance with SAE J607. A mechanical-type governor shall be provided for controlling the engine speed. Engines shall be so mounted that the air passages of the cooling system will not become clogged with cut grass thrown by the blades. Gas tank capacity shall be two quarts minimum.

3.5.4.1 Electromagnetic interference reduction. When specified (see 6.2), lawn mowers shall be equipped for electromagnetic interference suppression in accordance with MIL-STD-461 and tested (see 4.8) in accordance with SAE J551.

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3.5.4.2 Fungus proofing. When specified (see 6.2), lawn mowers, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components which are inherently fungus and water resistant or which are hermetically sealed need not be treated.

(b) Components whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to components or element surfaces previously cleaned and prepared in accordance with accepted commercial practice.

3.5.4.3 Safety switch. When specified (see 6.2), engines shall be equipped with a high temperature safety switch.

3.5.5 Controls. Mowers shall have controls designed for heavy duty use. The controls shall be positive in action and shall remain in the required position under all operating and handling conditions of the mower. Controls for reverse shall be of the deadman type so that power to the drive is automatically interrupted when the operator's actuating force is removed. Controls shall be conveniently located on the handle and within reach of the operator's normal position.

3.5.5.1 Size A machines. Size A machines shall be equipped with two controls. One control shall simultaneously regulate the engagement and disengagement of the cutting reel and traction drive, and the other control to regulate the engine speed (operation of the two controls by one rod or lever is acceptable). The supplier may, at his option, provide controls as specified in 3.4.5.2.

3.5.5.2 Size B machines. Size B machines shall have three controls for (a) regulating the engine speed, (b) engaging and disengaging the traction drive, and (c) engaging and disengaging the cutting reel drive. Each primary control shall operate independently of the other.

3.5.6 Handles. The handle frames shall be of metal construction and shall be sufficiently rigid to prevent whipping action or noticeable deformation when the mower is manipulated on turns or hills. Handles shall be formed from welded or seamless steel pipe or tubing. If the handle is in sections, it shall be welded or reinforced with bolted cross members to increase rigidity. Handle dimensions shall be not less than 7/8 of an inch outside diameter with not less than a 16 gage wall thickness. Hand grips shall be covered with rubber or other suitable composition which shall not slip when twisted by hand. Mower handles shall conform to the test requirements of 4.7.1.

3.5.7 Traction drive. The traction drive shall be by either wheel or roller unless a specific type is specified in the contract or order (see 6.2). Power to the driving wheels or rollers shall be transmitted through differential gearing. Power transmission between the engine and the differential shall be by roller chain and sprocket, gearshafts and gears, V-belts, or a combination thereof. Differential gears and transmission gears shall mesh fully and smoothly, shall be completely housed in a grease or oil filled case, and shall be suitable for heavy duty continuous use. Where a V-belt drive is used, a movable shaft and pulley or equally suitable device may be furnished to function as a clutch in engaging and disengaging the V-belt drive. V-belt drives shall be designed with either a take-up on the pulley or with an idler. Inside idlers shall be at least as large as the smallest loaded pulley. Outside idlers shall be one-third larger than the smallest loaded pulley. All pulleys used shall be cast iron, machined steel, or powdered metal (P/M) steel and securely mounted. All V-belts used shall be heavy duty belts, no fractional horsepower or light duty belts shall be used. V-belts used shall meet industry requirements for engine horsepower transmitted with a service factor of 1.5. At the time of bid submission, the contractor shall provide certification from the belt manufacturer, for his propulsion design, that all belts utilized on the unit are approved for their specific use for a minimum life of 500 hours.

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3.5.7.1 Wheel drive. Mowers with wheel drive shall have two drive wheels of metal construction of either the cast or disk type with disks welded, bolted, or riveted to each other. The wheels shall be fitted with solid, pneumatic, or semi-pneumatic rubber tires. Pneumatic rubber tires shall be of the industrial type, not less than two ply and not less than 3 inches nominal in sectional width. Solid and semi-pneumatic tires shall be not less than 2-1/2 inches nominal in width. Drive axles or wheels shall have wheel bearings of the ball or roller type packed in grease, either permanently sealed to prevent the entry of dirt and moisture or fitted with functional grease fittings to facilitate lubrication and the flushing out of dirt, etc.

3.5.7.2 Roller drive. Mowers with roller drive shall have a sectional metal drive roller, the overall length of which shall be approximately equal to the width of the cutting swath. All sections of the roller shall be power driven. Rollers shall be all metal or rubber covered metal. Rubber covered rollers shall have the rubber permanently molded to each section of the roller. Sufficient traction shall be obtainable to operate the lawn mower in conformity with the requirements specified in 3.4.6.

3.5.8 Propulsion capability. Mowers shall be capable of continuous cutting at a forward speed of 264 ft/minute on firm grass covered grounds. Mowers shall also be capable of continuous cutting directly up firm grades of 30 percent at a speed of at least 220 ft/minute. The reverse speed for size B mowers shall be a minimum of 140 ft/minute.

3.6 Grass catcher. When specified (see 6.2), mowers (other than models equipped with rear wheel casters) shall be furnished with grass catchers. They shall be either of sheet metal, woven wire, plastic, or metal-and-canvas construction. They shall be durable in service and shall be readily fastened to the lawn mower.

3.7 Cutting requirements. Rotating reel type mowers shall be capable of cutting heavy, thick-growth grass, 2-1/2 inches in height above the bed knife, in a single pass when operating under conditions herein specified. The cut shall be smooth and even for all heights of cut above the ground to which the bed knife is adjusted or elevated. The mowers shall not slip at the wheels nor mar the turf in normal operation.

3.7.1 Height of cut above ground. The height of cut shall be adjustable throughout a range of not less than 3/4 to 2 inches above the ground. The height of cut adjustment shall be firmly secured and shall not slip or get out of adjustment during the operation of the machine over rough terrain. The reel assembly shall not be tilted upon adjustment in a manner that will contribute to poor cutting alignment.

3.8 Sound pressure level. The sound pressure level shall not exceed 92 dbA, with the sound level meter set at slow response when tested in accordance with ANSI B71.1.

3.9 Lubrication. All components requiring lubrication shall be either permanently lubricated or shall be readily accessible for lubrication and provided with functional grease or oil fittings. The axle housing shall contain removable metal threaded fittings for providing adequate lubrication. The mower shall be lubricated when delivered.

3.10 Finish. All exterior metal surfaces shall be finished with a smooth durable weatherproof paint or lacquer to resist corrosion. Handle frames may be chromium or nickel plated.

3.11 Instructions and parts list information. The contractor shall furnish with each lawn mower either a booklet(s) or pamphlet(s) giving complete instructions for the operation, lubrication, adjustment, and care of the engine, mower, and attachment units. A pamphlet listing all repair parts and parts source information shall also be furnished with each lawn mower.

3.12 Identification marking. Each lawn mower shall be marked in a plain and permanent manner with the model designator and name of the manufacturer or with a trademark of such known character that the source of manufacturer may be readily determined.

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3.13 Workmanship. The completely fabricated mower shall conform to the quality and grade product covered by this specification. The mower shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, rust, deterioration, or malformations. There shall be no sharp edges, burrs, or slivers. Workmanship shall be first class throughout.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Bid sample inspection.
(b) Lot acceptance inspection.

4.3 Bid sample inspection. Bid sample inspection shall be applied to the lawn mowers submitted in accordance with 3.3. Failure of the bid sample to pass the examination of one or more bid sample tests shall be cause for rejection. The bid sample inspection shall consist of the examination in 4.6 and all the tests in 4.7.

4.4 Sampling for lot acceptance inspection.

4.4.1 Lot. All lawn mowers offered for delivery at one time shall be considered a lot for the purpose of acceptance, inspections, and tests.

4.4.2 Sampling for inspection. A random sample shall be selected from each lot in accordance with the provisions of MIL-STD-105 at inspection level II.

4.4.3 Sampling for tests. One sample lawn mower shall be selected at random. If the sample fails to comply with all of the tests, two additional samples shall be selected. If any additional sample fails any of the tests, mowers represented by the samples shall be considered as not conforming to the requirements of this specification. Manufacturers may correct minor deficiencies that cause failure.

4.5 Lot acceptance inspection. Lot acceptance inspection shall be applied to each item in accordance with 4.4.2 and 4.4.3 prior to being offered for acceptance under the contract. Lot acceptance inspection shall consist of the following:

(a) Examination (see 4.6).
(b) Tests (see 4.7.1 and 4.7.2).
(c) Inspection of preparation for delivery (see 4.10).

4.6 Examination.

4.6.1 Visual and dimensional examination. The completely assembled end item shall be examined for defects in finish, construction, workmanship, marking, and nonconformance with dimensional requirements. Defects shall be as classified in table I. The inspection level shall be level II with an Acceptable Quality Level (AQL) of 4.0 percent for major defects and 6.5 percent for total defects expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not finished where required.	X	
	Type of finish not as specified, or blistered, peeled, chipped, or area of no film or thin film.	X	
	Any component that is painted or otherwise coated where serviceability may be affected.	X	
	Significant rust.	X	
	Negligible rust.		X
Dimensions	Dimensions not in accordance with the specified requirements.	X	
Type	Not type specified.	X	
Size	Not size specified.	X	
Construction and workmanship, general	Part missing or not as specified.	X	
	Malformed, fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, or otherwise impaired.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel.	X	
	Component not readily accessible for servicing where required.		X
Welding or brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X	
	Slag inclusion, undercut, not smooth and uniform, scale, or flux deposits not removed.		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, broken, stripped, fractured.	X	
	Loose.		X
Assembly	Any component not properly assembled or secured.	X	
	Unit perceptibly out of square or alignment.	X	
Marking identification, instruction manual	Missing, incomplete, not legible, not as specified.	X	

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4.6.2 ANSI conformance. The manufacturer shall supply certification that the Mower(s) he intends to furnish the Government meets the safety requirements as specified in 3.4 and is capable of passing the ANSI tests.

4.7 Tests.

4.7.1 Handle assembly test. A torque load shall be applied to handle. The mower chassis shall be restrained during the test. The load shall be applied to the handle by means of a torque wrench coupled to a U-shaped channel iron type adapter (or equivalent) 12 inches long. The channel shall be made to straddle the handle with the torque wrench secured to the middle of the channel. The torque shall be applied in such a manner as to twist the handle about its center axis. An initial torque load of 50 foot-pounds shall be applied and then removed. The handle shall then be measured to determine its displacement from a plane parallel with the floor at a distance 6 inches from each side the center axis at which the torque was applied. This deformation shall be recorded. The final torque load of 100 foot-pounds shall then be applied and removed. Another measurement shall be made as described above and the difference in measurements shall not exceed 3/8 inch.

4.7.2 Frequency of cut. The frequency of cut shall be calculated from the number of knives on the reel, the circumference of the drive wheels, and the ratio of the number of revolutions made by the reel to the number of complete revolutions made by the drive wheels. The ratio may be determined from the number of revolutions of the reel for each revolution of the drive wheel. The frequency of cut shall be calculated as follows:

$$F = \frac{R \times N}{C}$$

Where:

F = Frequency (cuts per foot of travel).
R = Ratio of reel to wheel revolutions.
N = Number of reel knives.
C = Circumference of drive wheels (feet).

Mowers with a frequency of cut less than 9 shall be considered to have failed the frequency of cut requirement.

4.7.3 Inclined operation tests. Mowers shall be tested for stability and the engine's ability to operate on inclined positions. Mowers shall be placed on a variable slope, rectangular single plane (tilt table) at 25° to the horizontal. Each sample mower shall be tested in four positions: front and rear end and right and left side down hill, with the down hill wheels blocked to prevent slipping. Mowers shall then be started and operated at approximately 3/4 throttle for a period of two hours in each position or until the engine has failed from lack of lubrication or fuel. Engines shall not stall, consume, leak, or burn oil.

4.7.4 Propulsion capability test.

4.7.4.1 Level ground. The propulsion capability test shall be conducted on a straight, firm, fairly level course. Whenever practicable, the course shall contain heavy, thick grass or grass and weeds. The operator shall guide the mower toward a landmark or special marker. With the mower operating at the highest speed attainable and set at the minimum height adjustment, it shall be guided toward the landmark or marker while cutting for not less than two minutes to determine compliance with 3.5.8.

4.7.4.2 Grade. The grade propulsion test shall be conducted up and across, in both directions, on a grass course (see 4.7.4.1) with a grade of 30 percent to determine compliance with 3.5.8. The course shall run between two marked points not less than 55 feet apart. With the mower operating at the highest speed attainable, the time required (in seconds) for the mower to pass between the marked points shall be noted. The speed shall be calculated as follows:

$$\text{speed (ft/minute)} = \frac{60 \times \text{distance (feet)}}{\text{time (seconds)}}$$

Failure to attain the required speed for level ground or the 30 percent grade shall be cause for rejection.

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4.8 Radio interference suppression test. Conformance of the sample lawn mower with MIL-STD-461 will be determined by authorized representatives of the contracting officer using the test method set forth in SAE J551 and Government test equipment. The testing facility will furnish the contracting officer a report on the sample tested, including details of the suppression system applicable for production units and recommendations for quality control of production. Upon approval by the contracting officer and provided all other requirements of this specification are met, the contractor shall examine each production unit to determine that it conforms to the report of the contracting officer and the approved suppression system of the tested sample.

4.9 Rejected lots. Rejected lots may be resubmitted for examination and tests, provided the contractor has removed or repaired all nonconforming lawn mowers. Samples shall again be selected and inspected from such resubmitted lots to verify compliance with this specification. Rejected lawn mowers shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections and measures taken to overcome the defects.

4.10 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing, and marking requirements are in compliance with section 5 of this specification. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 percent defects per hundred units in accordance with MIL-STD-105.

TABLE II. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior and interior containers)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as: Incomplete closure of container, loose strapping, inadequate stapling. Bulging or distortion of containers.
Preservation	Improperly applied or missing.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Disassembly. Parts vulnerable to damage, pilferage, and loss or that would otherwise increase cubage, such as controls, control rods, levers, and handles shall be removed. All bolts, nuts, pins, and washers removed shall be placed in mating parts and secured to prevent their loss. Small parts removed shall be match-marked to facilitate reassembly. Match-marking information shall be on cloth tags conforming to UU-T-81, type A.

5.1.1.2 Unprotected metal surfaces. Unprotected metal surfaces requiring the application of a contact preservative shall be coated with a type P-1 preservative conforming to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, and the following exceptions and additional requirements.

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5.1.1.3.1 Exterior surfaces. Exterior surfaces of the engine and engine accessories shall not be sprayed with insulation compound specified therein.

5.1.1.3.2 Wheels and axles. Wheels removed shall have the bearing surfaces of the axle or spindles, interior bearing surfaces of the hub, and all ball or roller bearings which are not removed from the axles or spindles coated with type P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A and the barrier material secured in place with tape conforming to PPP-T-60, class 1.

5.1.1.3.3 Drive wheels. Drive wheels shall be removed from the mowers and the interior surfaces, gears, and drive control clutches coated with a lubricant conforming to MIL-L-2105 and the wheels replaced.

5.1.1.3.4 Drive chains and sprockets. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration to the inner surfaces of the rollers, pins, and bushings. After excess type P-3 preservative has drained, the entire chain shall be coated with type P-1 preservative. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.1.1.3.5 Drive belts and pulleys. Pulley faces or grooves shall be coated with primer conforming to TT-P-664. The drive belts shall remain in place on the equipment and shall have the tension relieved.

5.1.1.3.6 Gears and gear housings. When the lubricant contained in the gear housing conforms to the specified operating lubricant, MIL-L-2105 or MIL-L-21260, a quantity of the applicable lubricant required to raise the lubricant to the operating level in the housing shall be added. The gears shall be actuated to insure coating all interior parts and surfaces. The threads of the fill plugs shall be coated with lubricant conforming to MIL-L-21260 before the plugs are reinstalled.

5.1.1.3.7 Clutches. Metal components of clutches shall be coated with primer conforming to TT-P-664.

5.1.1.3.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.3.9 Technical publications. Technical publications shall be preserved and packaged in accordance with MIL-P-116, Method IC-1.

5.1.2 Level C. Mowers, parts, tools, and technical publications shall be preserved in a manner which will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use or controlled humidity storage.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Each complete mower, parts, tools, and technical publications shall be packed together in a close fitting box conforming to PPP-B-621, class 2, style optional, or PPP-B-601, overseas type, style optional, or PPP-B-640, class 2. Blocking, bracing, or cushioning shall be provided to prevent the disassembled parts from coming in contact with each other. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Each complete mower, parts, tools, and technical publications shall be packed together as specified in 5.2.1, except that the boxes shall be class 1 domestic type. Alternatively, power lawn mowers with a cut of 22 inches or less shall be packed in a fiberboard box conforming to PPP-B-636, class domestic, variety DW, grade 350. The box shall be fitted with internal corner posts or a liner of fiberboard, grade not less than 275.

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5.2.3 Level C. Each complete mower shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the requirements on the Uniform Freight Classification or National Motor Freight Classification, as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or purchase order (see 6.2), the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The mowers covered by this specification are intended for heavy-duty commercial use in the care and maintenance of lawns and grounds for which the Government is responsible.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size (see 1.2.1).
(c) Bid sample required (see 3.3).
(d) Electromagnetic interference reduction, if required (see 3.4.4.1).
(e) Fungus proofing, if required (see 3.4.4.2).
(f) High temperature safety switch, if required (see 3.4.4.3).
(g) When either wheel or roller drive is required (see 3.4.7).
(h) When grass catcher is required (see 3.5).
(i) Width of cut, if different (see 3.5).
(j) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).
(k) Special marking, if required (see 5.3).

6.3 Definition of terms used in specification. The term lawn mowers as used herein refer to hand-pushed, selfpropelled machines having power-driven cutting devices which produce cutting swaths of the size specified in 1.1. This specification also covers the grass catchers (see 3.5) which may be attached to the appropriate type of lawn mowers.

6.4 Supersession data. When finalized, this specification is intended to supersede type I, class 2, style 1, size A and type I, class 2, style 2, size D of 00-M-00681c(GSA-FSS) dated April 30, 1965, and type I, class B, grade A, sizes A and D or 00-M-681a dated August 30, 1955.

6.5 Service guarantee. It is suggested that the purchasing officer include the following clause in the invitation for bids and all awarded contracts:

All necessary adjustments of machines, with the exception of engines which are subject to the warrantees of their respective manufacturers, procured hereunder, not occasioned by misuse or accident through fault or negligence by the Government, shall be made by the contractor at his own expense, including transportation costs, if any, during the 90-day period after the machine is put in service. All mowers procured hereunder are guaranteed for a period of six months from the date the machine is put in service. During the guarantee period, all broken or defective parts not caused by accident or misuse through fault or negligence by the Government, must be replaced (including labor, transportation, and parts) at the contractor's expense. Also, all necessary machine adjustments occasioned by such defective parts must be made at the contractor's expense, including transportation costs, if any. This guarantee also applies to breakage of mower parts caused by objects thrown by the cutting knives. The contractor agrees to have all parts for the item furnished available for a period of five years from the date of award, and agrees to bill the Government for spare parts not covered by the warranty at their industry-wide published list price. The contractor agrees that parts, during this five year period of availability, will be shipped immediately. If the part is not in stock at the time the order is received, it will be shipped within not more than thirty days after receipt of order.

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User activities:

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Navy - MC

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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